

E-Waste Robotics Battery Extraction Project

Dataset Collection and Annotation Report

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Project Title

Android Phone Battery Detection Dataset for E-Waste Robotics Battery Extraction

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1. Introduction

Electronic waste (e-waste) is rapidly becoming one of the fastest-growing environmental problems in the modern world. Mobile phones play a major role in this issue because millions of devices are thrown away each year. One of the most critical and dangerous parts inside smartphones is the lithium-ion battery. Improper handling of batteries during recycling can lead to environmental pollution, fire hazards, and safety dangers.

The objective behind this project is to contribute towards the creation of an intelligent robotics mechanism that can detect and extract batteries from abandoned Android handsets. In order to accomplish the goal, it became imperative to develop a custom computer vision dataset for detecting batteries in smartphones.

This dataset consists of images of batteries from various Android smartphones that have been appropriately annotated through bounding boxes in order to identify their location in the image.

2. Project Objectives

The main objectives of this dataset creation project are:

- To collect high-quality images of Android phone batteries.
- To prepare a clean and diverse image dataset suitable for computer vision tasks.
- To annotate battery regions accurately using bounding boxes.
- To create a reusable dataset for robotic battery extraction research.
- To publish the dataset on Kaggle for future academic and research use.

3. Dataset Overview

Item	Description
Dataset Name	Android Phone Battery Detection Dataset
Total Images	125 Images
Annotation Type	Bounding Box Annotation
Annotation Tool	CVAT.ai
Image Sources	YouTube repair videos and the iFixit website
Dataset Purpose	Battery detection for robotic e-waste extraction
Image Processing Tools	Microsoft Designer
Output Format	Object Detection Dataset
Upload Platform	Kaggle

The dataset contains images of Android phone internal components with visible batteries. Each image includes annotations identifying the exact battery location using bounding boxes.

4. Data Collection Process

4.1 Image Collection Strategy

During the data collection process, emphasis was placed on capturing realistic pictures of Android phone interiors in which batteries are visible because the project is based on robotic extraction in e-waste systems.

Data collection was done through;

1. YouTube smartphone repairing videos
2. Ifixit website Android phone repairing sections

This source presented numerous Android phones in various configurations in terms of;

1. Phone models
2. Battery types
3. Lighting settings
4. Camera views
5. Battery positions

4.2 Collecting Images from YouTube Videos

The repair videos from YouTube were chosen as one of the major sources of imagery since they depict practical repair situations. Phone disassembly and battery replacement videos were thus chosen.

Steps Undertaken

1. Searched for Android phone repair and battery replacement videos.
2. Watched videos frame by frame to determine the presence of battery visibility.
3. Taken screenshots depicting the inner battery construction.
4. Saved the screenshots in respective folders.
5. Deleted duplicate and out-of-focus images.

Benefits of Utilizing YouTube Videos

- Present practical repair settings.
- Offer various viewing angles.
- Illustrate varying conditions of batteries.

Screenshot Evidence Placeholder

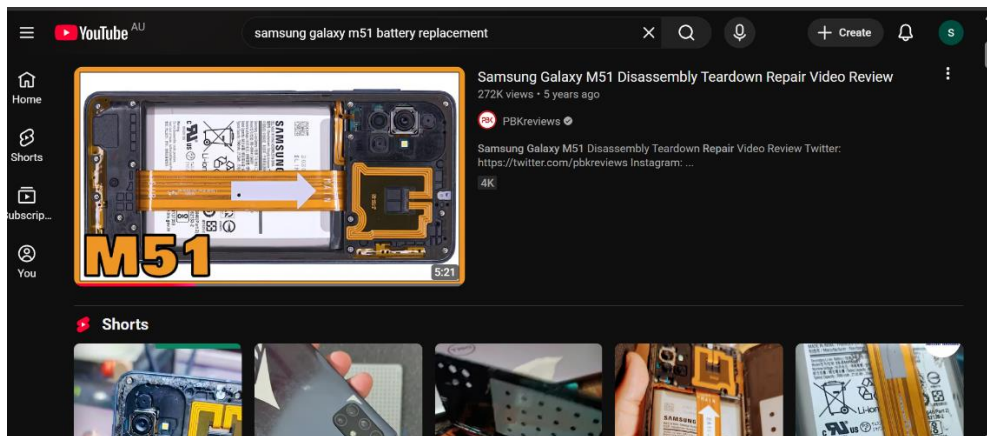


Figure 01: YouTube video source selection

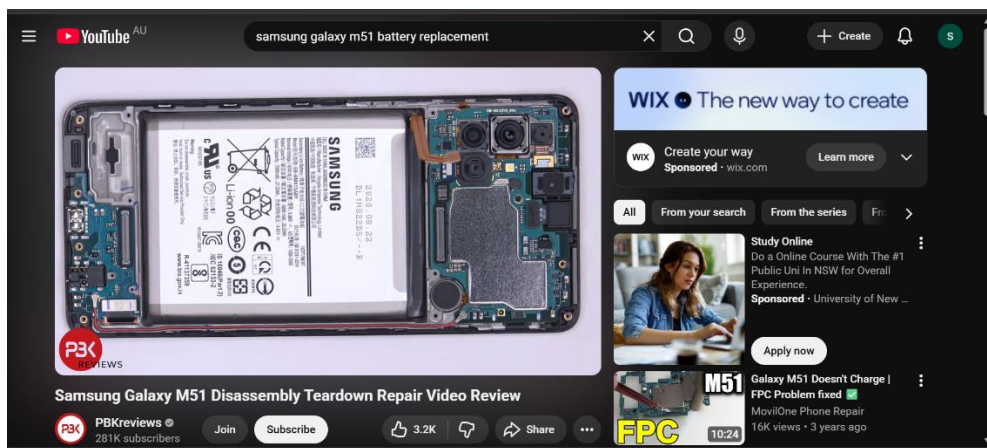


Figure 02: Screenshot capture process from YouTube video

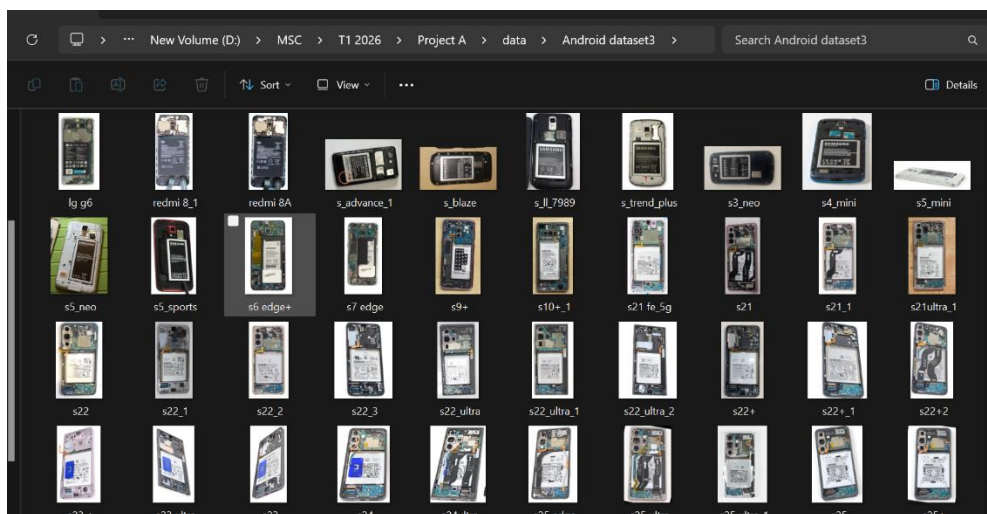


Figure 03: Folder containing collected screenshots

4.3 Collecting Images from the iFixit Website

Also employed for obtaining image resources from the internet was the iFixit website due to the availability of repair manuals that provide clear images.

Website URL:

https://www.ifixit.com/Device/Android_Phone

Procedure Followed

1. Browsed Android phone repair manuals.
2. Searched through teardown instructions.
3. Picked images showing batteries very clearly.
4. Taken screenshots.
5. Saved images in classified folders.

Benefits Associated with Using the iFixit Website

- High-quality images.
- Professional repair manual.
- Good visibility of components in the phone.
- Availability in various Android devices.

Screenshot Evidence Placeholder

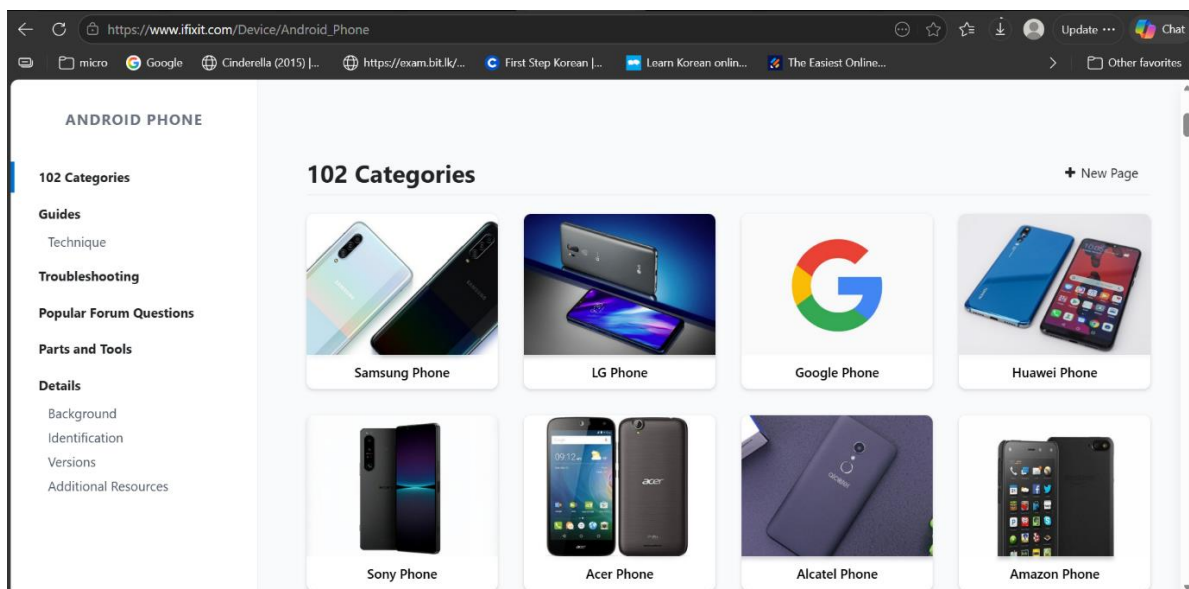


Figure 04: iFixit Android device page

Step 15 Loudspeaker Assembly

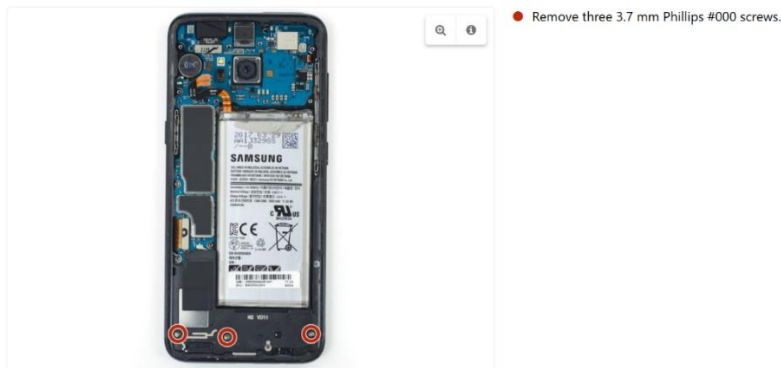


Figure 05: Example teardown image with visible battery

5. Source Platforms That Were Used

5.1 YouTube

YouTube provided much of the source material for dynamic visuals related to repairs. Careful selection was made to ensure the battery is visible in these videos.

Example Content Collected

- Battery changing videos
- Disassembly videos
- Repairs videos
- Inspection videos

5.2 iFixit

The iFixit platform provides structured and professional repair guides for electronic devices. The Android phone section was particularly useful for obtaining clean and detailed battery images.

Benefits

- Standardized repair steps
- Consistent image quality
- Multiple device models
- Detailed internal layouts

6. Image Preparation and Cleaning

After collecting the screenshots, image preprocessing and cleaning were completed to improve the dataset quality.

The Microsoft Designer application was used to edit and enhance the images.

6.1 Objectives of Image Cleaning

The image cleaning process aimed to:

- Improve visibility of batteries.
- Remove unnecessary background objects.
- Align images consistently.
- Enhance image quality.
- Create uniform training data.

6.2 Editing Operations Performed

Cropping

Images were cropped to remove unnecessary areas and focus on the phone's internals.

Rotation

Images were rotated to correct orientation issues.

Alignment Adjustments

Battery positions were aligned properly to improve annotation accuracy.

Background Removal/Change

Unwanted backgrounds were removed or minimized to make battery areas more visible.

Screenshot Evidence Placeholder



Figure 06: Original unedited image

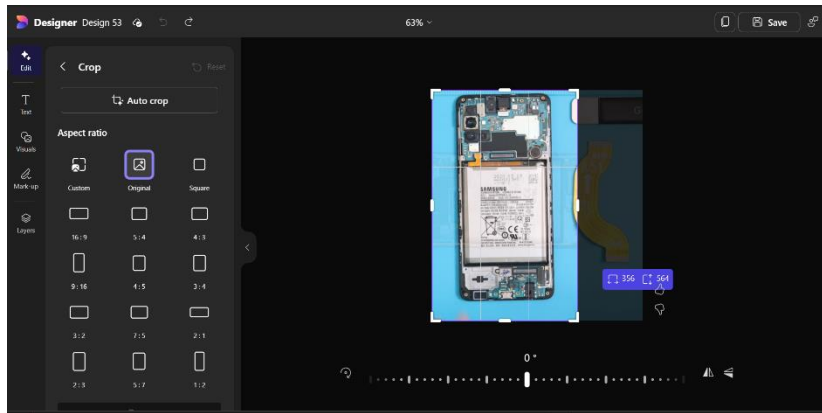


Figure 07: Cropping process in Microsoft Designer

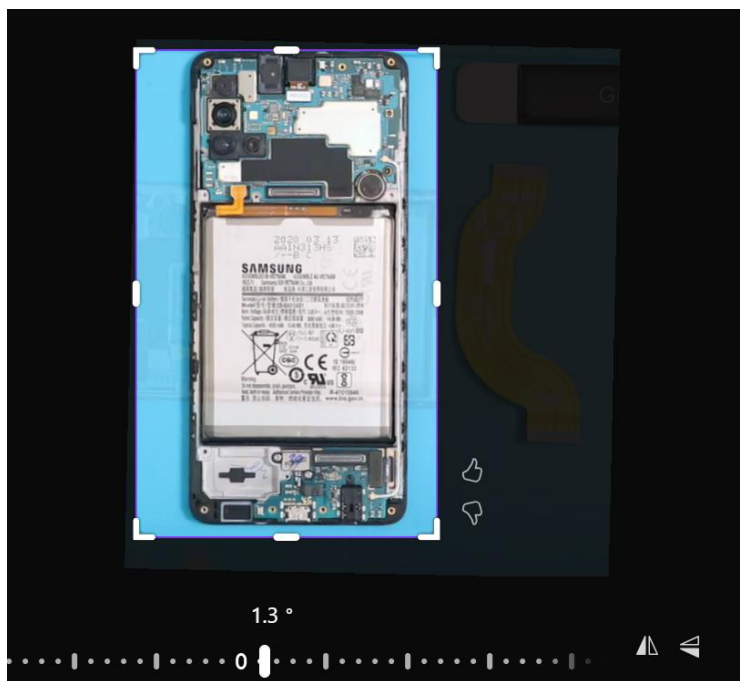


Figure 08: Rotation and alignment adjustment

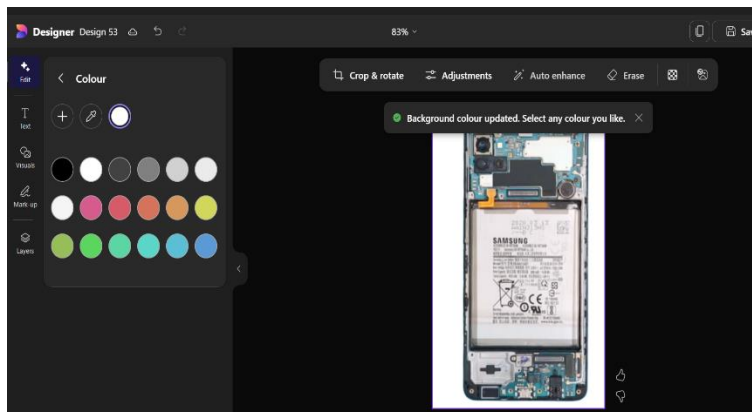


Figure 09: Background removal/editing process



Figure 10: Final cleaned image

6.3 Importance of Data Cleaning

Proper image preparation improves machine learning performance by reducing noise and inconsistencies in the dataset. Clean images help object detection models identify the target object more accurately.

The cleaning process also helped standardize image quality across different data sources.

7. Annotation Process Using CVAT.ai

7.1 Introduction to CVAT.ai

CVAT.ai is an open-source annotation platform commonly used for computer vision tasks such as object detection, segmentation, and tracking.

For this project, CVAT.ai was used to annotate battery regions using bounding boxes.

7.2 Annotation Workflow

Step 1 – Creating a New Task

A new annotation task was created in CVAT.ai for the battery detection dataset.

Step 2 – Uploading Images

The cleaned images were uploaded into CVAT.ai.

Step 3 – Creating Labels

A label named “Battery” was created for annotation.

Step 4 – Drawing Bounding Boxes

Bounding boxes were manually drawn around each visible battery.

Step 5 – Reviewing Annotations

Each annotation was checked for accuracy and consistency.

Step 6 – Exporting Dataset

The annotated dataset was exported in object detection format for future model training.

7.3 Bounding Box Annotation

Bounding box annotation was selected because:

- It is suitable for object detection models.
- It provides accurate object localization.
- It is computationally efficient.
- It works effectively with YOLO-based architectures.

Each battery was enclosed tightly within a rectangular box to improve detection precision.

7.4 Annotation Quality Assurance

Several quality checks were performed:

- Verified all images contained annotations.
- Ensured bounding boxes fully enclosed batteries.
- Corrected overlapping or inaccurate boxes.
- Removed incorrect annotations.
- Maintained annotation consistency.

7.5 Screenshot Evidence Placeholder

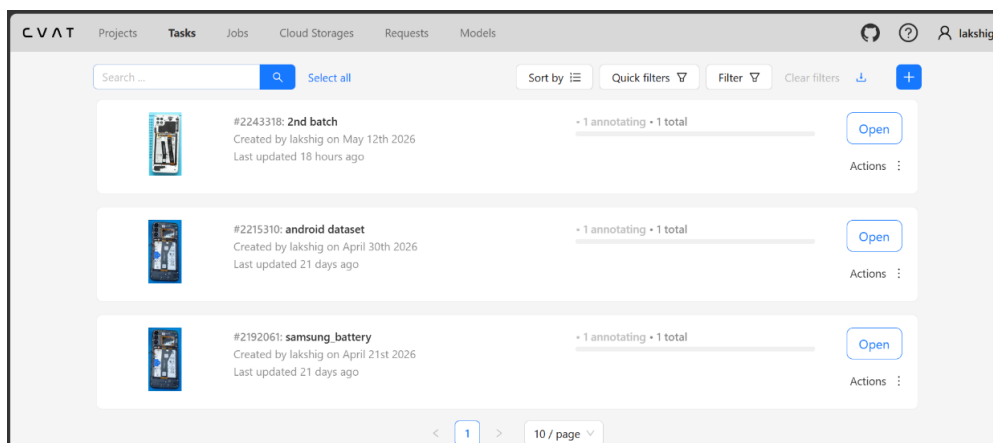


Figure 11: CVAT.ai task page

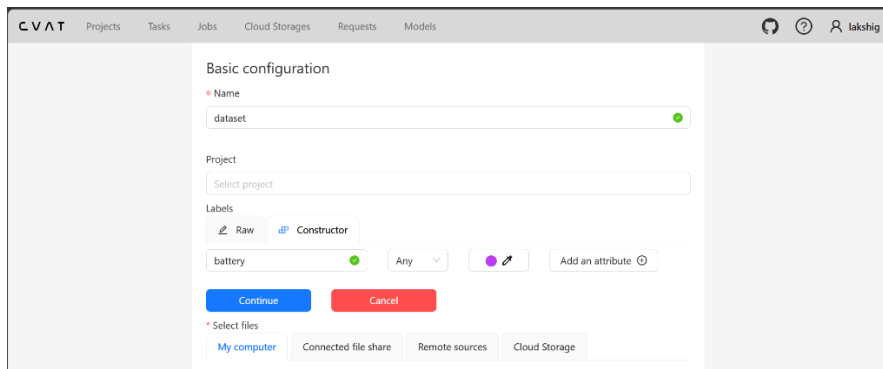


Figure 12: Task creation page

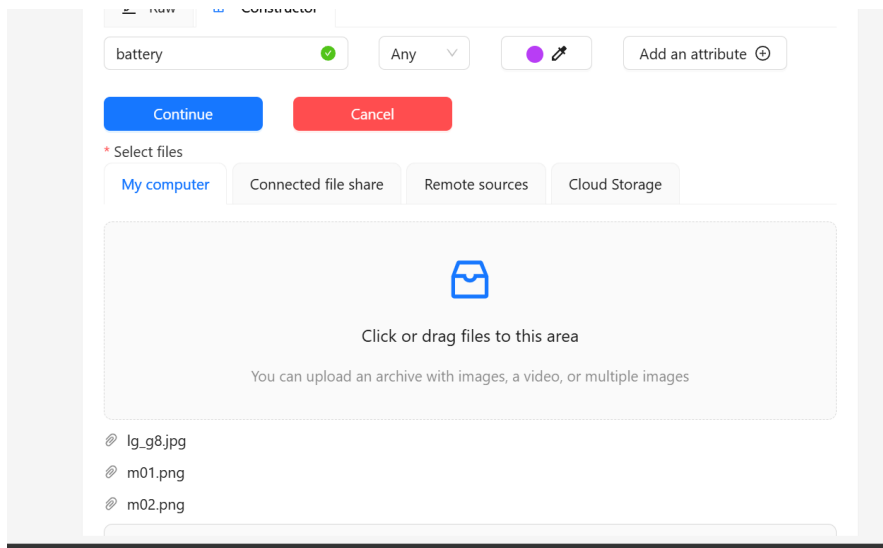


Figure 13: Image upload process

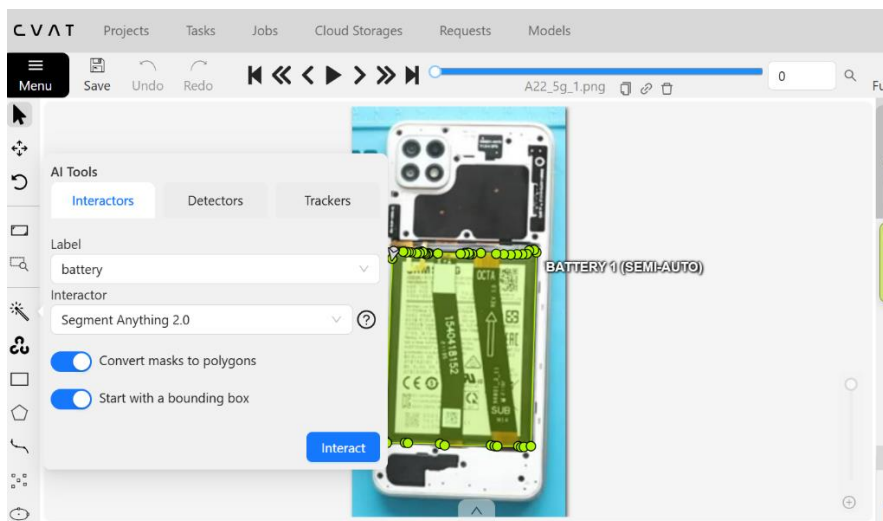


Figure 14: Bounding box annotation example

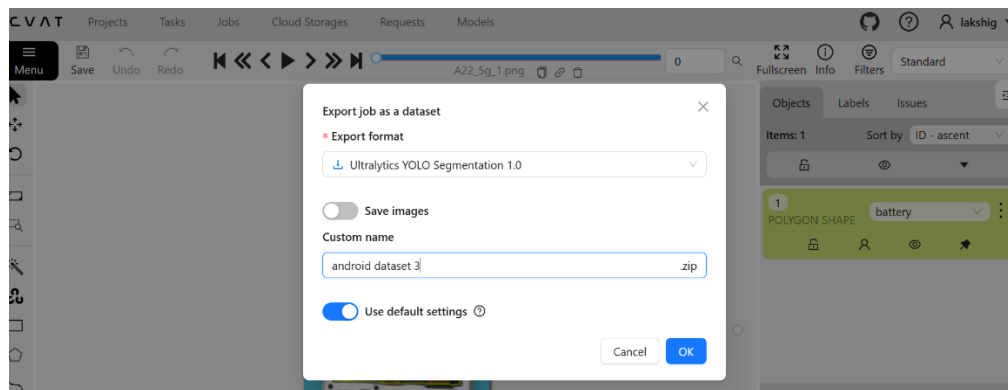


Figure 15: Dataset export process

8. Dataset Organization

The dataset was organized into structured folders for easier management and future model training.

8.1 Folder Structure

```
Android_Dataset/
├── image_001.png
├── image_002.png
├── image_003.png
├── ...
├──
├── Android_dataset/
│   ├── train.txt
│   ├── data.yaml
│   ├── labels/
│   │   └── train.txt
```

9. Quality Assurance and Verification

To ensure dataset reliability, several verification procedures were followed.

9.1 Image Verification

- Checked image clarity.
- Removed duplicates.
- Removed low-quality screenshots.
- Ensured battery visibility.

9.2 Annotation Verification

- Reviewed all bounding boxes.
- Verified label consistency.
- Corrected annotation placement errors.

9.3 Dataset Consistency

- Maintained consistent annotation standards.
- Standardized image preparation procedures.
- Organized file naming structures.

10. Dataset Upload to Kaggle

After completing image annotation and verification, the dataset was uploaded to Kaggle for future access and research use.

10.1 Purpose of Uploading to Kaggle

- Share the dataset with researchers.
- Enable future machine learning experiments.
- Support reproducibility.
- Preserve the dataset online.

10.2 Kaggle Upload Process

Steps Followed

1. Organized dataset folders.
2. Uploaded images and annotation files.
3. Added dataset description.
4. Added metadata and tags.
5. Published the dataset.

10.3 Screenshot Evidence Placeholder

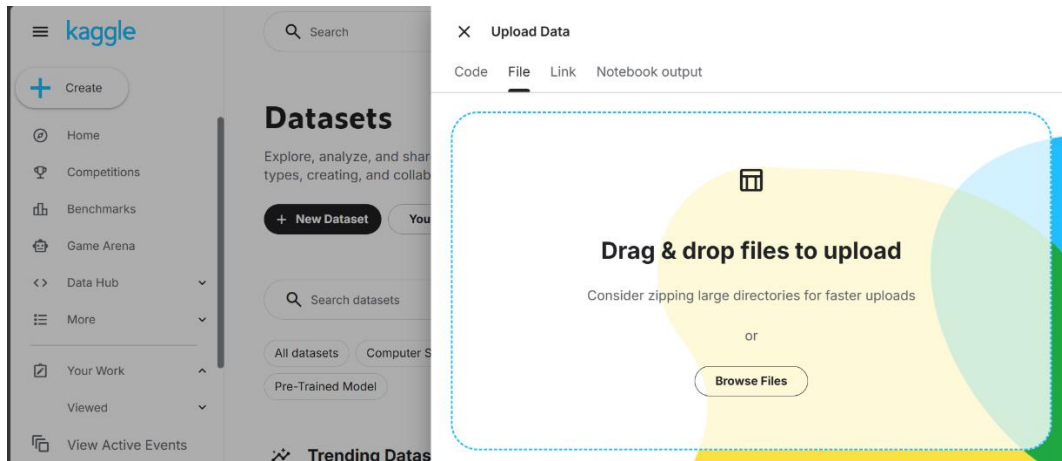


Figure 16: Kaggle dataset creation page

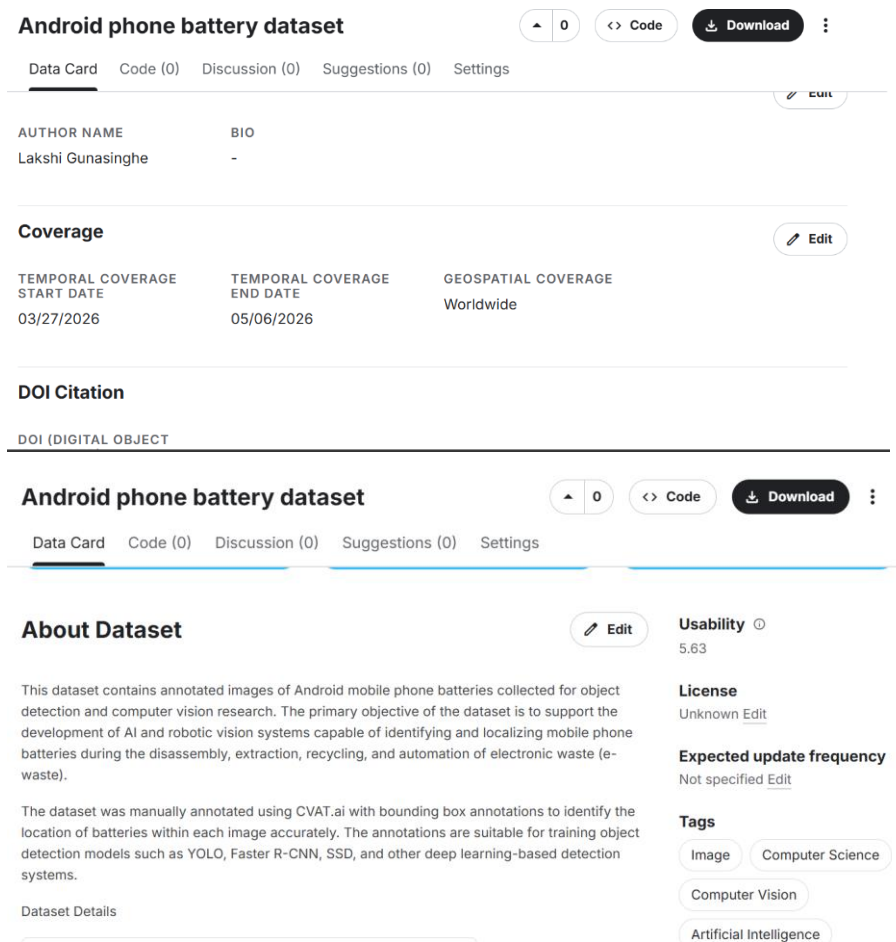


Figure 17: Dataset metadata and description

<https://www.kaggle.com/datasets/lluckymun/android-phone-battery-dataset>

Link 01: Published Kaggle dataset page

11. Challenges Faced

Several challenges were encountered during the dataset creation process.

11.1 Image Quality Variations

Images collected from videos had different resolutions and lighting conditions.

Solution

Image cleaning and enhancement were performed using Microsoft Designer.

11.2 Different Battery Shapes and Sizes

Android phones contain batteries with varying shapes and placements.

Solution

A diverse image collection strategy was used to improve dataset variability.

11.3 Annotation Time Consumption

Manual bounding box annotation required significant time and attention.

Solution

Annotations were reviewed systematically to maintain accuracy.

11.4 Background Complexity

Some images contained distracting repair tools and internal phone components.

Solution

Background cleanup and image cropping were applied.

12. Future Improvements

The dataset can be improved further in future work.

Possible Improvements

- Increase dataset size.
- Include more smartphone brands.
- Add segmentation annotations.
- Include damaged battery samples.
- Capture real industrial e-waste scenarios.
- Add robotic arm interaction images.
- Expand to multi-object annotations.

13. Conclusion

This project successfully created a custom Android phone battery detection dataset useful for e-waste robotic battery extraction research.

In total, there were 125 images gathered from various YouTube repair video tutorials and the iFixit Android Phone Repair website. These images were cleaned and enhanced using Microsoft Designer.

The custom dataset was then labeled using CVAT.ai, which involved creating bounding boxes around the areas containing batteries within the images. Quality assurance testing was conducted at different stages during the labeling phase.

Lastly, the dataset was uploaded to Kaggle.

With a custom dataset available, we can now use deep learning algorithms to train an object detector to detect smartphone batteries.

14. References

1. YouTube repair and teardown videos [Samsung Galaxy M51 Disassembly Teardown Repair Video Review - YouTube](#)
2. iFixit Android Device Repair Guides [Android Phone Repair Help: Learn How to Fix It Yourself.](#)
3. CVAT.ai Annotation Platform [Leading Data Annotation Platform for Images, Videos and 3D | CVAT](#)
4. Microsoft Designer Application
5. Kaggle Dataset Platform [Kaggle: The World's AI Proving Ground](#)